

Exponentials & Logarithms

Disclaimer: This pdf can be used or distributed for non-profit purpose. It is against owners will that you use this for profit making or any other source(s) of income.

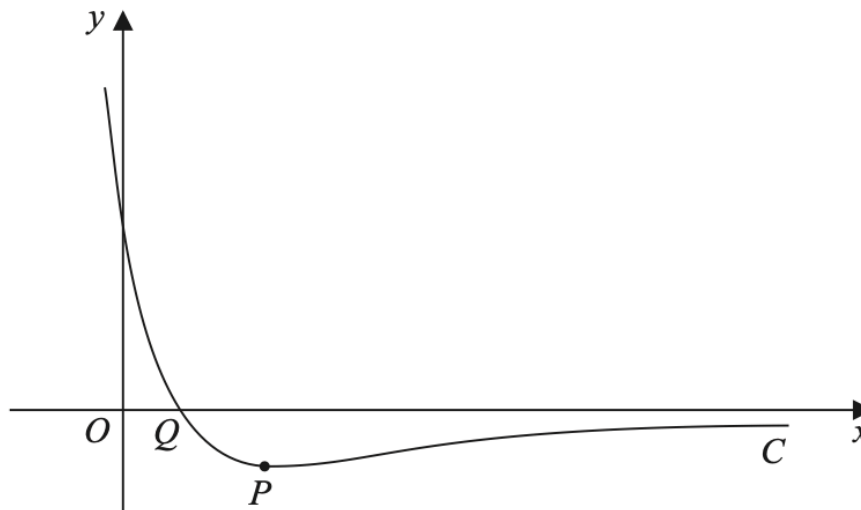
June 2014 Q11 C34**11.****Figure 2**

Figure 2 shows a sketch of part of the curve C with equation

$$y = e^{a-3x} - 3e^{-x}, \quad x \in \mathbb{R}$$

where a is a constant and $a > \ln 4$

The curve C has a turning point P and crosses the x -axis at the point Q as shown in Figure 2.

(a) Find, in terms of a , the coordinates of the point P .

(6)

(b) Find, in terms of a , the x coordinate of the point Q .

(3)

(c) Sketch the curve with equation

$$y = |e^{a-3x} - 3e^{-x}|, \quad x \in \mathbb{R}, \quad a > \ln 4$$

Show on your sketch the exact coordinates, in terms of a , of the points at which the curve meets or cuts the coordinate axes.

(3)

8.

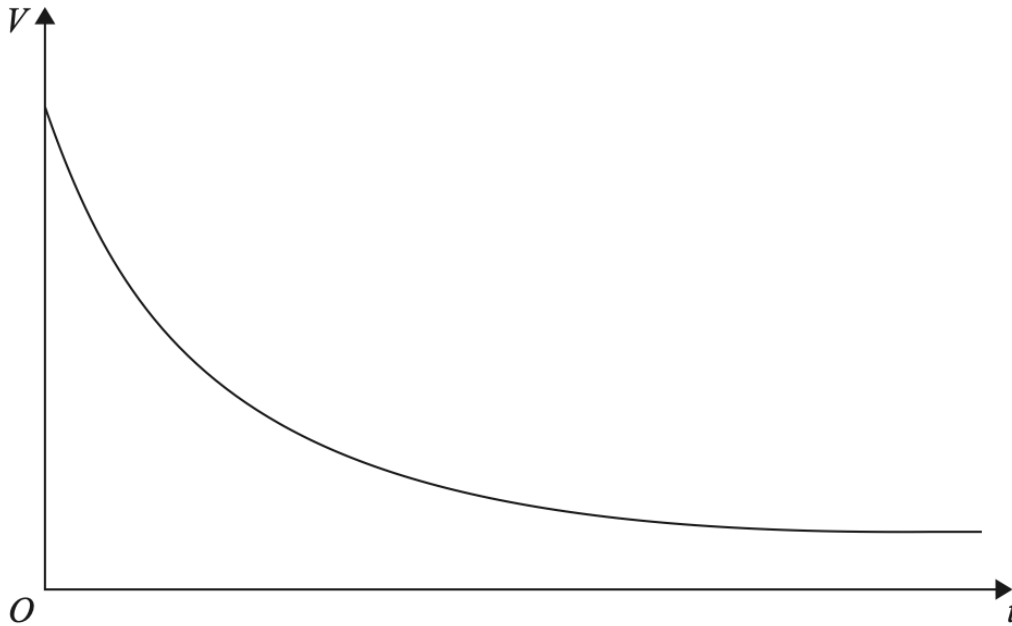


Figure 1

The value of Lin's car is modelled by the formula

$$V = 18000e^{-0.2t} + 4000e^{-0.1t} + 1000, \quad t \geq 0$$

where the value of the car is V pounds when the age of the car is t years.

A sketch of t against V is shown in Figure 1.

(a) State the range of V .

(2)

According to this model,

(b) find the rate at which the value of the car is decreasing when $t = 10$
Give your answer in pounds per year.

(3)

(c) Calculate the exact value of t when $V = 15000$

(4)

June 2015 Q6 C34

6. The mass, m grams, of a radioactive substance t years after first being observed, is modelled by the equation

$$m = 25e^{1-kt}$$

where k is a positive constant.

- (a) State the value of m when the radioactive substance was first observed.

(1)

Given that the mass is 50 grams, 10 years after first being observed,

- (b) show that $k = \frac{1}{10} \ln\left(\frac{1}{2}e\right)$

(4)

- (c) Find the value of t when $m = 20$, giving your answer to the nearest year.

(3)

January 2016 Q6 C34

6. A hot piece of metal is dropped into a cool liquid. As the metal cools, its temperature T degrees Celsius, t minutes after it enters the liquid, is modelled by

$$T = 300e^{-0.04t} + 20, \quad t \geq 0$$

- (a) Find the temperature of the piece of metal as it enters the liquid.

(1)

- (b) Find the value of t for which $T = 180$, giving your answer to 3 significant figures.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

- (c) Show, by differentiation, that the rate, in degrees Celsius per minute, at which the temperature of the metal is changing, is given by the expression

$$\frac{20 - T}{25}$$

(3)

9.

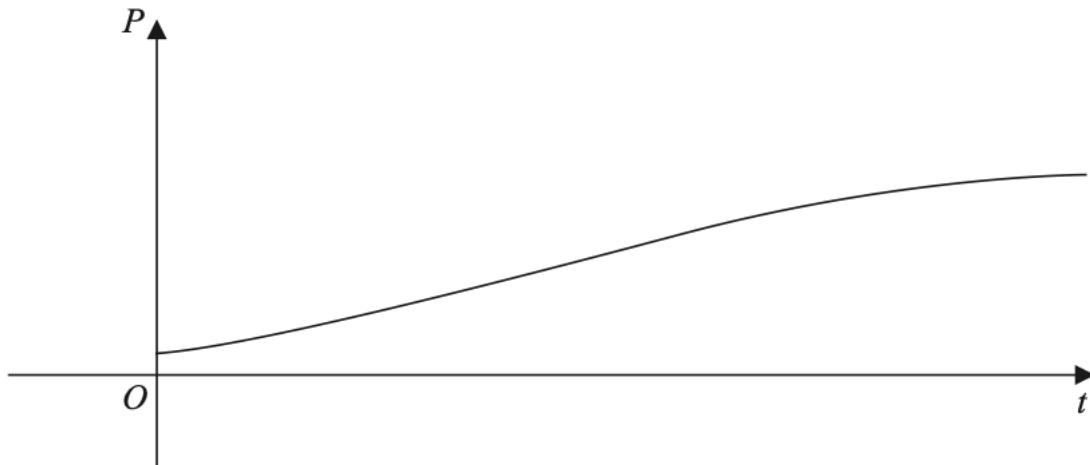


Figure 2

The population of a species of animal is being studied. The population P , at time t years from the start of the study, is assumed to be

$$P = \frac{9000e^{kt}}{3e^{kt} + 7}, \quad t \geq 0$$

where k is a positive constant.

A sketch of the graph of P against t is shown in Figure 2.

Use the given equation to

(a) find the population at the start of the study, (2)

(b) find the value for the upper limit of the population. (1)

Given that $P = 2500$ when $t = 4$

(c) calculate the value of k , giving your answer to 3 decimal places. (5)

Using this value for k ,

(d) find, using $\frac{dP}{dt}$, the rate at which the population is increasing when $t = 10$

Give your answer to the nearest integer.

(3)

January 2017 Q10 C34

10. A population of insects is being studied. The number of insects, N , in the population, is modelled by the equation

$$N = \frac{300}{3 + 17e^{-0.2t}} \quad t \in \mathbb{R}, t \geq 0$$

where t is the time, in weeks, from the start of the study.

Using the model,

- (a) find the number of insects at the start of the study, (1)
- (b) find the number of insects when $t = 10$, (2)
- (c) find the time from the start of the study when there are 82 insects.
(Solutions based entirely on graphical or numerical methods are not acceptable.) (4)
- (d) Find, by differentiating, the rate, measured in insects per week, at which the number of insects is increasing when $t = 5$. Give your answer to the nearest whole number. (3)

13.

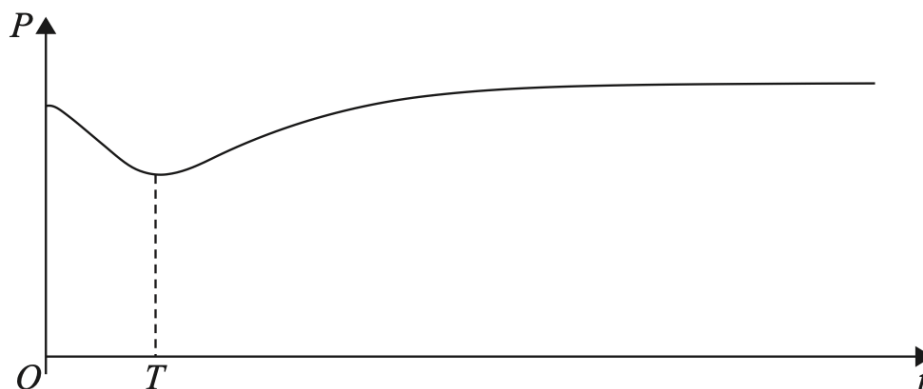


Figure 5

A colony of ants is being studied. The number of ants in the colony is modelled by the equation

$$P = 200 - \frac{160e^{0.6t}}{15 + e^{0.8t}} \quad t \in \mathbb{R}, t \geq 0$$

where P is the number of ants, measured in thousands, t years after the study started. A sketch of the graph of P against t is shown in Figure 5

- (a) Calculate the number of ants in the colony at the start of the study. (2)
- (b) Find $\frac{dP}{dt}$ (3)

The population of ants initially decreases, reaching a minimum value after T years, as shown in Figure 5

- (c) Using your answer to part (b), calculate the value of T to 2 decimal places.

(Solutions based entirely on graphical or numerical methods are not acceptable.) (4)

October 2017 Q3 C34

3. The number of bacteria in a liquid culture is modelled by the formula

$$N = 3500(1.035)^t, \quad t \geq 0$$

where N is the number of bacteria t hours after the start of a scientific study.

- (a) State the number of bacteria at the start of the scientific study. (1)
- (b) Find the time taken from the start of the study for the number of bacteria to reach 10 000
Give your answer in hours and minutes, to the nearest minute. (4)
- (c) Use calculus to find the rate of increase in the number of bacteria when $t = 8$
Give your answer, in bacteria per hour, to the nearest whole number. (3)

October 2018 Q9 C34

9. A rare species of mammal is being studied. The population P , t years after the study started, is modelled by the formula

$$P = \frac{900e^{\frac{1}{4}t}}{3e^{\frac{1}{4}t} - 1}, \quad t \in \mathbb{R}, \quad t \geq 0$$

Using the model,

- (a) calculate the number of mammals at the start of the study,

(1)

- (b) calculate the exact value of t when $P = 315$

Give your answer in the form $a \ln k$, where a and k are integers to be determined.

(4)

- (c) (i) Find $\frac{dP}{dt}$

- (ii) Hence find the value of $\frac{dP}{dt}$ when $t = 8$, giving your answer to 2 decimal places.

(4)

January 2019 Q13 C34

13. A scientist is studying a population of insects. The number of insects, N , in the population, t days after the start of the study is modelled by the equation

$$N = \frac{240}{1 + ke^{\frac{t}{16}}}$$

where k is a constant.

Given that there were 50 insects at the start of the study,

- (a) find the value of k (2)

- (b) use the model to find the value of t when $N = 100$ (3)

- (c) Show that

$$\frac{dN}{dt} = \frac{1}{p}N - \frac{1}{q}N^2$$

where p and q are integers to be found.

(5)

June 2019 Q5 C34

5. A bath is filled with hot water. The temperature, $\theta^\circ\text{C}$, of the water in the bath, t minutes after the bath has been filled, is given by

$$\theta = 20 + Ae^{-kt}$$

where A and k are positive constants.

Given that the temperature of the water in the bath is initially 38°C ,

- (a) find the value of A .

(2)

The temperature of the water in the bath 16 minutes after the bath has been filled is 24.5°C .

- (b) Show that $k = \frac{1}{8} \ln 2$

(4)

Using the values for k and A ,

- (c) find the temperature of the water 40 minutes after the bath has been filled, giving your answer to 3 significant figures.

(2)

- (d) Explain why the temperature of the water in the bath cannot fall to 19°C .

(1)

14.

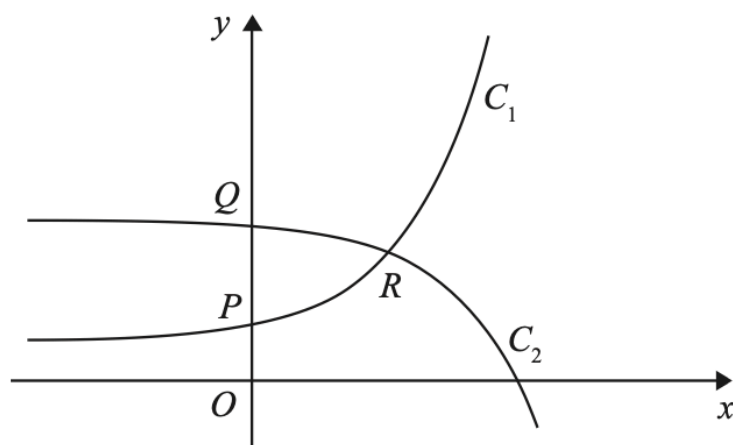


Figure 5

Figure 5 shows a sketch of the curves C_1 and C_2

C_1 has equation $y = 3 + e^{x+1}$ $x \in \mathbb{R}$

C_2 has equation $y = 10 - e^x$ $x \in \mathbb{R}$

Given that C_1 and C_2 cut the y -axis at the points P and Q respectively,

(a) find the exact distance PQ .

(2)

C_1 and C_2 intersect at the point R .

(b) Find the exact coordinates of R .

(5)

January 2020 Q1 P3

1. A population of a rare species of toad is being studied.

The number of toads, N , in the population, t years after the start of the study, is modelled by the equation

$$N = \frac{900e^{0.12t}}{2e^{0.12t} + 1} \quad t \geq 0, t \in \mathbb{R}$$

According to this model,

- (a) calculate the number of toads in the population at the start of the study, (1)
- (b) find the value of t when there are 420 toads in the population, giving your answer to 2 decimal places. (4)
- (c) Explain why, according to this model, the number of toads in the population can never reach 500 (1)

3.

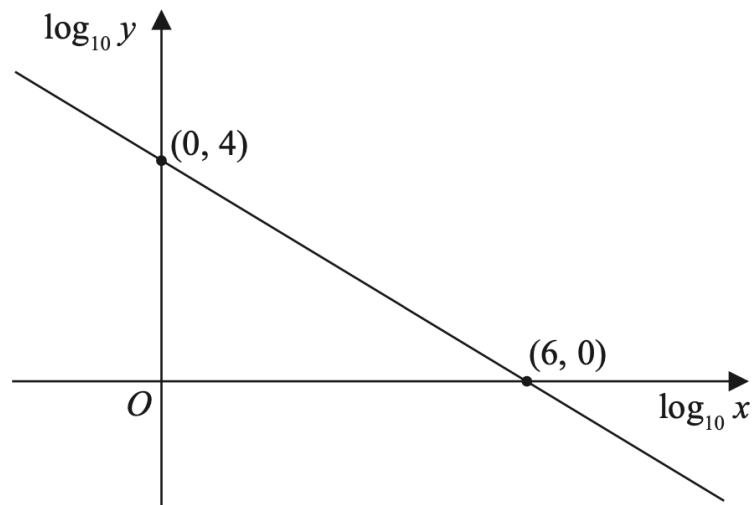


Figure 1

Figure 1 shows a linear relationship between $\log_{10} y$ and $\log_{10} x$

The line passes through the points $(0, 4)$ and $(6, 0)$ as shown.

- (a) Find an equation linking $\log_{10} y$ with $\log_{10} x$ (2)
- (b) Hence, or otherwise, express y in the form px^q , where p and q are constants to be found. (3)

October 2020 Q2 P3

2. A scientist monitored the growth of bacteria on a dish over a 30-day period.

The area, $N\text{mm}^2$, of the dish covered by bacteria, t days after monitoring began, is modelled by the equation

$$\log_{10} N = 0.0646t + 1.478 \quad 0 \leq t \leq 30$$

- (a) Show that this equation may be written in the form

$$N = ab^t$$

where a and b are constants to be found. Give the value of a to the nearest integer and give the value of b to 3 significant figures.

(4)

- (b) Use the model to find the area of the dish covered by bacteria 30 days after monitoring began. Give your answer, in mm^2 , to 2 significant figures.

(2)

January 2021 Q5 P3

5. The temperature, $\theta^{\circ}\text{C}$, inside an oven, t minutes after the oven is switched on, is given by

$$\theta = A - 180e^{-kt}$$

where A and k are positive constants.

Given that the temperature inside the oven is initially 18°C ,

- (a) find the value of A .

(2)

The temperature inside the oven, 5 minutes after the oven is switched on, is 90°C .

- (b) Show that $k = p \ln q$ where p and q are rational numbers to be found.

(4)

Hence find

- (c) the temperature inside the oven 9 minutes after the oven is switched on, giving your answer to 3 significant figures,

(2)

- (d) the rate of increase of the temperature inside the oven 9 minutes after the oven is switched on. Give your answer in $^{\circ}\text{C min}^{-1}$ to 3 significant figures.

(3)

January 2021 Q8 P3

8. The percentage, P , of the population of a small country who have access to the internet, is modelled by the equation

$$P = ab^t$$

where a and b are constants and t is the number of years after the start of 2005

Using the data for the years between the start of 2005 and the start of 2010, a graph is plotted of $\log_{10} P$ against t .

The points are found to lie approximately on a straight line with gradient 0.09 and intercept 0.68 on the $\log_{10} P$ axis.

- (a) Find, according to the model, the value of a and the value of b , giving your answers to 2 decimal places. (4)
- (b) In the context of the model, give a practical interpretation of the constant a . (1)
- (c) Use the model to estimate the percentage of the population who had access to the internet at the start of 2015 (2)

5.

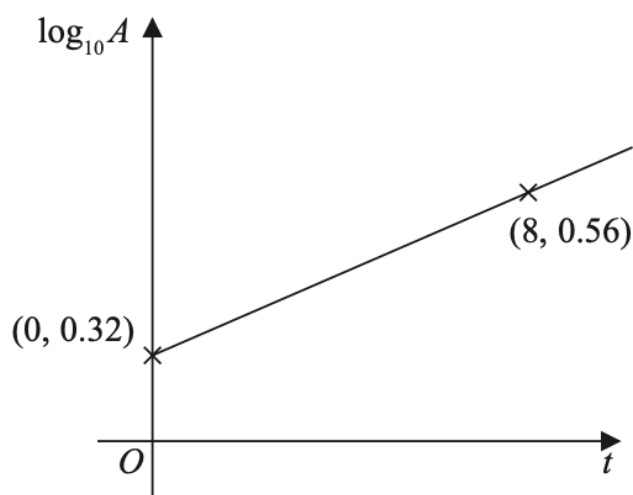


Figure 1

The growth of duckweed on a pond is being studied.

The surface area of the pond covered by duckweed, $A\text{m}^2$, at a time t days after the start of the study is modelled by the equation

$$A = pq^t \quad \text{where } p \text{ and } q \text{ are positive constants}$$

Figure 1 shows the linear relationship between $\log_{10} A$ and t .

The points $(0, 0.32)$ and $(8, 0.56)$ lie on the line as shown.

(a) Find, to 3 decimal places, the value of p and the value of q .

(4)

Using the model with the values of p and q found in part (a),

(b) find the rate of increase of the surface area of the pond covered by duckweed, in m^2/day , exactly 6 days after the start of the study.
Give your answer to 2 decimal places.

(3)

June 2021 Q8 P3

8. A scientist is studying a population of fish in a lake. The number of fish, N , in the population, t years after the start of the study, is modelled by the equation

$$N = \frac{600e^{0.3t}}{2 + e^{0.3t}} \quad t \geq 0$$

Use the equation of the model to answer parts (a), (b), (c), (d) and (e).

- (a) Find the number of fish in the lake at the start of the study. (1)
- (b) Find the upper limit to the number of fish in the lake. (1)
- (c) Find the time, after the start of the study, when there are predicted to be 500 fish in the lake. Give your answer in years and months to the nearest month. (4)
- (d) Show that

$$\frac{dN}{dt} = \frac{Ae^{0.3t}}{(2 + e^{0.3t})^2}$$

where A is a constant to be found.

(3)

Given that when $t = T$, $\frac{dN}{dt} = 8$

- (e) find the value of T to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

3.

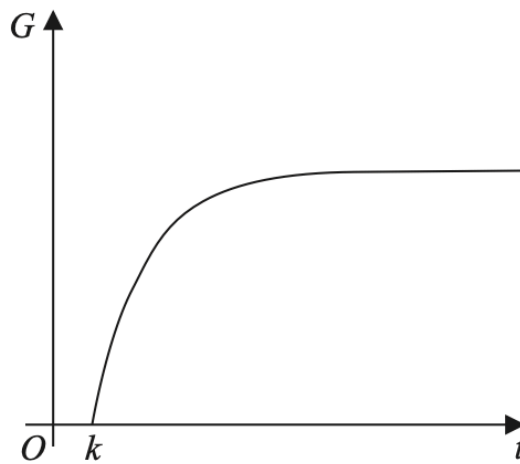


Figure 2

The total mass of gold, G tonnes, extracted from a mine is modelled by the equation

$$G = 40 - 30e^{1-0.05t} \quad t \geq k \quad G \geq 0$$

where t is the number of years after 1st January 1800.

Figure 2 shows a sketch of G against t .

Use the equation of the model to answer parts (a), (b) and (c).

(a) (i) Find the value of k .

(ii) Hence find the year and month in which gold started being extracted from the mine.

(3)

(b) Find the total mass of gold extracted from the mine up to 1st January 1870.

(2)

There is a limit to the mass of gold that can be extracted from the mine.

(c) State the value of this limit.

(1)

October 2021 Q7 P3

- 7 The mass, M kg, of a species of tree can be modelled by the equation

$$\log_{10} M = 1.93 \log_{10} r + 0.684$$

where r cm is the base radius of the tree.

The base radius of a particular tree of this species is 45 cm.

According to the model,

- (a) find the mass of this tree, giving your answer to 2 significant figures.

(2)

- (b) Show that the equation of the model can be written in the form

$$M = pr^q$$

giving the values of the constants p and q to 3 significant figures.

(3)

- (c) With reference to the model, interpret the value of the constant p .

(1)

January 2022 Q4 P3

4. The growth of a weed on the surface of a pond is being studied.

The surface area of the pond covered by the weed, $A \text{ m}^2$, is modelled by the equation

$$A = \frac{80pe^{0.15t}}{pe^{0.15t} + 4}$$

where p is a positive constant and t is the number of days after the start of the study.

Given that

- 30 m^2 of the surface of the pond was covered by the weed at the start of the study
- 50 m^2 of the surface of the pond was covered by the weed T days after the start of the study

- (a) show that $p = 2.4$

(2)

- (b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on graphical or numerical methods are not acceptable.)

(4)

The weed grows until it covers the surface of the pond.

- (c) Find, according to the model, the maximum possible surface area of the pond.

(1)

January 2022 Q8 P3

8. A dose of antibiotics is given to a patient.

The amount of the antibiotic, x milligrams, in the patient's bloodstream t hours after the dose was given, is found to satisfy the equation

$$\log_{10} x = 2.74 - 0.079t$$

- (a) Show that this equation can be written in the form

$$x = pq^{-t}$$

where p and q are constants to be found. Give the value of p to the nearest whole number and the value of q to 2 significant figures.

(4)

- (b) With reference to the equation in part (a), interpret the value of the constant p .

(1)

When a different dose of the antibiotic is given to another patient, the values of x and t satisfy the equation

$$x = 400 \times 1.4^{-t}$$

- (c) Use calculus to find, to 2 significant figures, the value of $\frac{dx}{dt}$ when $t = 5$

(3)

4.

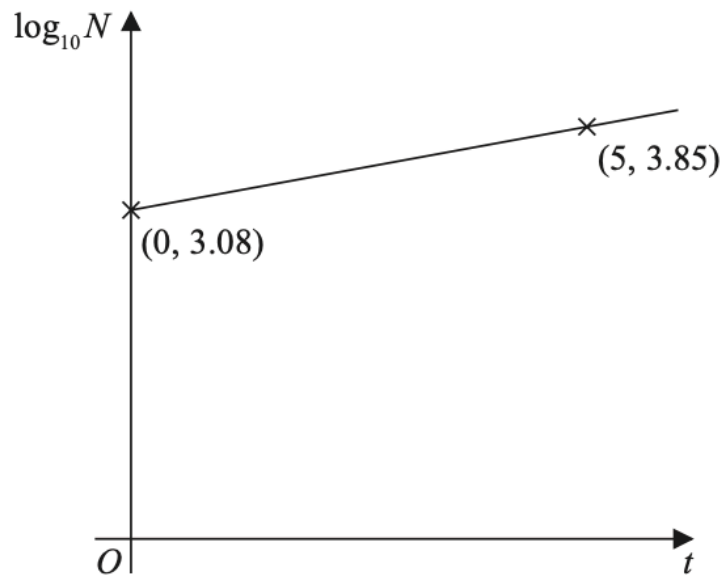


Figure 1

The number of subscribers to an online video streaming service, N , is modelled by the equation

$$N = ab^t$$

where a and b are constants and t is the number of years since monitoring began.

The line in Figure 1 shows the linear relationship between t and $\log_{10} N$

The line passes through the points $(0, 3.08)$ and $(5, 3.85)$

Using this information,

(a) find an equation for this line. (2)

(b) Find the value of a and the value of b , giving your answers to 3 significant figures. (3)

When $t = T$ the number of subscribers is 500 000

According to the model,

(c) find the value of T (2)

8.

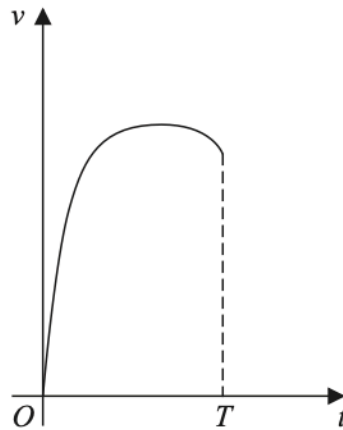


Figure 4

Figure 4 is a graph showing the velocity of a sprinter during a 100 m race.

The sprinter's velocity during the race, $v \text{ m s}^{-1}$, is modelled by the equation

$$v = 12 - e^{t-10} - 12e^{-0.75t} \quad t \geq 0$$

where t seconds is the time after the sprinter begins to run.

According to the model,

- (a) find, using calculus, the sprinter's maximum velocity during the race. (5)

Given that the sprinter runs 100 m in T seconds, such that

$$\int_0^T v \, dt = 100$$

- (b) show that T is a solution of the equation

$$T = \frac{1}{12} (116 - 16e^{-0.75T} + e^{T-10} - e^{-10}) \quad (4)$$

The iteration formula

$$T_{n+1} = \frac{1}{12} (116 - 16e^{-0.75T_n} + e^{T_n-10} - e^{-10})$$

is used to find an approximate value for T

Using this iteration formula with $T_1 = 10$

- (c) find, to 4 decimal places,
- (i) the value of T_2
 - (ii) the time taken by the sprinter to run the race, according to the model. (3)

October 2022 Q4 P3

4.

$$y = \log_{10}(2x + 1)$$

(a) Express x in terms of y .

(2)

(b) Hence, giving your answer in terms of x , find $\frac{dy}{dx}$

(3)

5.

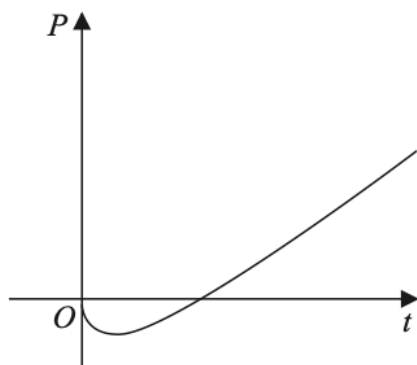


Figure 2

The profit made by a company, £ P million, t years after the company started trading, is modelled by the equation

$$P = \frac{4t-1}{10} + \frac{3}{4} \ln \left[\frac{t+1}{(2t+1)^2} \right]$$

The graph of P against t is shown in Figure 2.

According to the model,

- (a) show that exactly one year after it started trading, the company had made a loss of approximately £830 000 (2)

A manager of the company wants to know the value of t for which $P = 0$

- (b) Show that this value of t occurs in the interval $[6, 7]$ (2)
- (c) Show that the equation $P = 0$ can be expressed in the form

$$t = \frac{1}{4} + \frac{15}{8} \ln \left[\frac{(2t+1)^2}{t+1} \right] \quad (2)$$

- (d) Using the iteration formula

$$t_{n+1} = \frac{1}{4} + \frac{15}{8} \ln \left[\frac{(2t_n+1)^2}{t_n+1} \right] \quad \text{with } t_1 = 6$$

find the value of t_2 and the value of t_6 , giving your answers to 3 decimal places. (3)

- (e) Hence find, according to the model, how many months it takes in total, from when the company started trading, for it to make a profit. (2)

3.

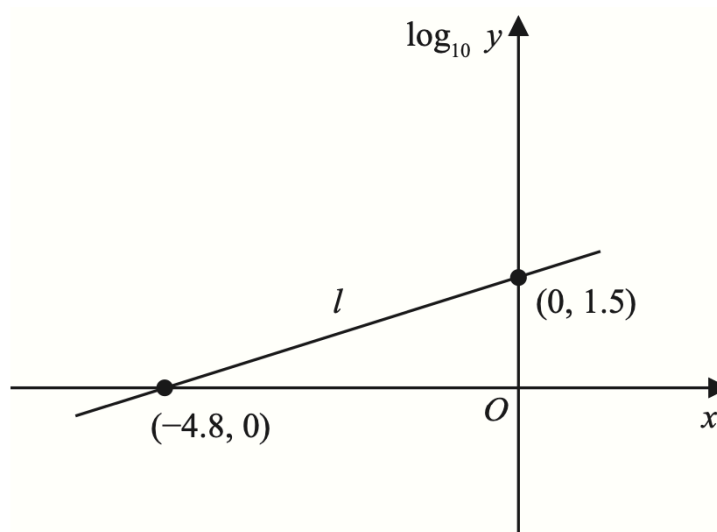


Figure 1

The line l in Figure 1 shows a linear relationship between $\log_{10} y$ and x .

The line passes through the points $(0, 1.5)$ and $(-4.8, 0)$ as shown.

(a) Write down an equation for l .

(2)

(b) Hence, or otherwise, express y in the form kb^x , giving the values of the constants k and b to 3 significant figures.

(3)

January 2023 Q10 P3

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A population of fruit flies is being studied.

The number of fruit flies, F , in the population, t days after the start of the study, is modelled by the equation

$$F = \frac{350e^{kt}}{9 + e^{kt}}$$

where k is a constant.

Use the equation of the model to answer parts (a), (b) and (c).

(a) Find the number of fruit flies in the population at the start of the study.

(1)

Given that there are 200 fruit flies in the population 15 days after the start of the study,

(b) show that $k = \frac{1}{15} \ln 12$

(3)

Given also that, when $t = T$, the number of fruit flies in the population is increasing at a rate of 10 per day,

(c) find the possible values of T , giving your answers to one decimal place.

(5)

2.

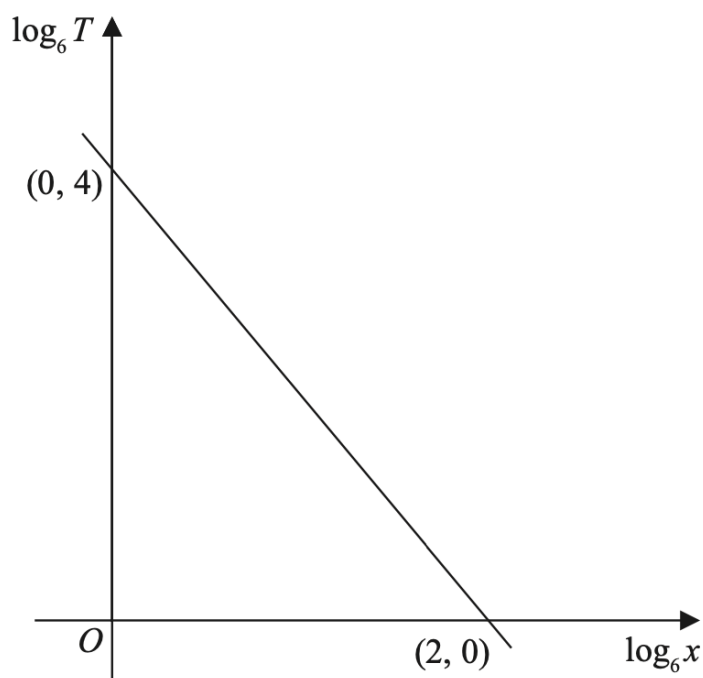


Figure 1

Figure 1 shows the linear relationship between $\log_6 T$ and $\log_6 x$

The line passes through the points $(0, 4)$ and $(2, 0)$ as shown.

(a) (i) Find an equation linking $\log_6 T$ and $\log_6 x$

(ii) Hence find the exact value of T when $x = 216$

(3)

(b) Find an equation, not involving logs, linking T with x

(3)

June 2023 Q7 P3

7. A scientist is studying two different populations of bacteria.

The number of bacteria N in the first population is modelled by the equation

$$N = Ae^{kt} \quad t \geq 0$$

where A and k are positive constants and t is the time in hours from the start of the study.

Given that

- there were 2500 bacteria in this population at the start of the study
- there were 10 000 bacteria 8 hours later

(a) find the exact value of A and the value of k to 4 significant figures.

(3)

The number of bacteria N in the second population is modelled by the equation

$$N = 60\,000e^{-0.6t} \quad t \geq 0$$

where t is the time in hours from the start of the study.

(b) Find the rate of decrease of bacteria in this population exactly 5 hours from the start of the study. Give your answer to 3 significant figures.

(2)

When $t = T$, the number of bacteria in the two different populations was the same.

(c) Find the value of T , giving your answer to 3 significant figures.

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

October 2023 Q4 P3

4. A new mobile phone is released for sale.

The total sales N of this phone, in **thousands**, is modelled by the equation

$$N = 125 - Ae^{-0.109t} \quad t \geq 0$$

where A is a constant and t is the time in months after the phone was released for sale.

Given that when $t = 0$, $N = 32$

- (a) state the value of A .

(1)

Given that when $t = T$ the total sales of the phone was 100 000

- (b) find, according to the model, the value of T . Give your answer to 2 decimal places.

(3)

- (c) Find, according to the model, the rate of increase in total sales when $t = 7$, giving your answer to 3 significant figures.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)

The total sales of the mobile phone is expected to reach 150 000

Using this information,

- (d) give a reason why the given equation is not suitable for modelling the total sales of the phone.

(1)

October 2023 Q6 P3

6. An area of sea floor is being monitored.

The area of the sea floor, $S \text{ km}^2$, covered by coral reefs is modelled by the equation

$$S = pq^t$$

where p and q are constants and t is the number of years after monitoring began.

Given that

$$\log_{10} S = 4.5 - 0.006t$$

- (a) find, according to the model, the area of sea floor covered by coral reefs when $t = 2$ (2)
- (b) find a complete equation for the model in the form

$$S = pq^t$$

giving the value of p and the value of q each to 3 significant figures.

(3)

- (c) With reference to the model, interpret the value of the constant q

(1)

January 2024 Q3 P3

3. The amount of money raised for a charity is being monitored.

The total amount raised in the t months after monitoring began, $\pounds D$, is modelled by the equation

$$\log_{10} D = 1.04 + 0.38t$$

- (a) Write this equation in the form

$$D = ab^t$$

where a and b are constants to be found. Give each value to 4 significant figures.

(3)

When $t = T$, the total amount of money raised is $\pounds 45\,000$

According to the model,

- (b) find the value of T , giving your answer to 3 significant figures.

(2)

The charity aims to raise a total of $\pounds 350\,000$ within the first 12 months of monitoring.

According to the model,

- (c) determine whether or not the charity will achieve its aim.

(2)

January 2024 Q5 P3

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The temperature, T °C, of the air in a room t minutes after a heat source is switched off, is modelled by the equation

$$T = 10 + Ae^{-Bt}$$

where A and B are constants.

Given that the temperature of the air in the room at the instant the heat source was switched off was 18 °C,

(a) find the value of A (1)

Given also that, exactly 45 minutes after the heat source was switched off, the temperature of the air in the room was 16 °C,

(b) find the value of B to 3 significant figures. (3)

Using the values for A and B ,

(c) find, according to the model, the rate of change of the temperature of the air in the room exactly two minutes after the heat source was switched off.
Give your answer in °C min⁻¹ to 3 significant figures. (2)

(d) Explain why, according to the model, the temperature of the air in the room cannot fall to 5 °C (1)